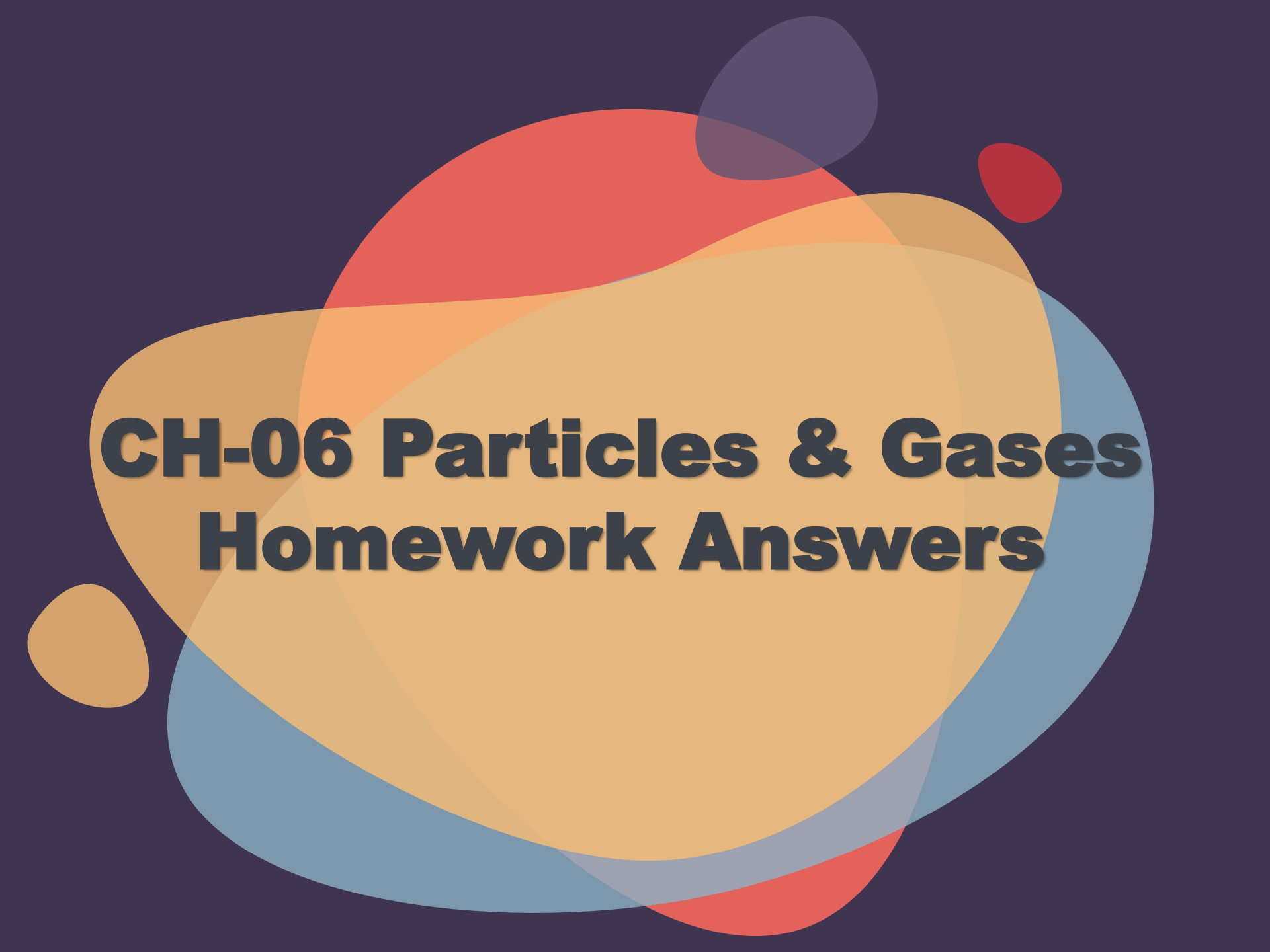




CH-06 Particles & Gases Homework Answers



01

A worker's exposure to an airborne contaminant in any 8-hour work shift of any 40-hour work in a week is known as the:

A. Threshold limit value.

B. Time weighted average.

C. Short term exposure limit.

D. Permissible exposure limit.



02

You are preparing a solution of 100ml of hydrogen with air equaling a volume of 1m^3 . What is the concentration of hydrogen in the mixture?

A. 1000 ppm

B. 10 ppm

C. 100 ppm

D. 1 ppm

$$\frac{100\text{ml} \times 10^{-3}}{1000} = 100\text{ml} \times 10^{-6}$$

$$(100\text{ml} \times 10^{-6}) \times 10^6 = 100 \text{ ppm}$$



03

An air sampling report shows the concentration of benzene in a personal air sample was 223 mg/m³. The molecular weight of benzene is 78.11 g/mol. What is this value expressed in parts per million?

A. 50 ppm

B. 70 ppm

C. 90 ppm

D. 110 ppm

$$ppm = \frac{\frac{mg}{m^3} \times 24.45}{MW}$$

$$ppm = \frac{223 \times 24.45}{78.11} = 70 ppm$$



04

Calculate the time weighted average for the following sample results taken from a worker in one shift.

8:00 am to 10:30 am 22ppm

10:30 am to 12:30 pm 12ppm

12:30 pm to 4:00 pm 2 ppm

A. 11 ppm

B. 12 ppm

C. 17 ppm

D. 22 ppm

$$TWA = \frac{(C_1T_1) + (C_2T_2) + \dots (C_nT_n)}{\text{total hours workes}}$$

$$TWA = \frac{22 \times 2.5 + 12 \times 2 + 2 \times 3.5}{2.5 + 2 + 3.5} = \frac{86}{8} = 10.75 \text{ ppm}$$



05

The OSHA level of airborne asbestos fibers allowed over a time-weighted average of 480 minutes is currently limited to 0.1 fiber per cubic centimeter. What is this limit called?

A. STEL

B. PEL

C. EL

D. REL

06

A cylinder with a volume of 4 ft^3 contains helium gas in a laboratory. The cylinder gauge reads 1500 psi. A technician opens the valve of the cylinder and releases helium directly into the laboratory until the gauge reads 1485 psi. What is the volume of the helium in the room?

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad \text{as } T_1 = T_2 \quad \text{so } P_1 V_1 = P_2 V_2$$

A. 0.04 ft^3 B. 0.41 ft^3 C. 4.1 ft^3 D. 41 ft^3

1

$$P_1 = 1,500 \text{ psi}$$

$$P_2 = 1500 - 1485 = 15 \text{ psi}$$

$$V_1 = 4 \text{ ft}^3$$

$$V_2 = ??$$

$$V_2 = \frac{15 \times 4}{1500} = 0.04 \text{ ft}^3$$

2

$$P_1 = 1,500 \text{ psi}$$

$$P_2 = \text{atmospheric pressure} = 14.7 \text{ psi}$$

$$V_1 = 0.04 \text{ ft}^3$$

$$V_2 = ??$$

$$V_2 = \frac{1500 \times 0.04}{14.7} = 4.1 \text{ ft}^3$$



07

An industrial hygienist collected the following personal dust samples. Is the employee's exposure above the OSHA PEL of 15 mg/m³?

Sample	Time Minutes	Flow Rate LPM	Result mg/m ³
1	120	2.0	12.05
2	240	1.98	5.65
3	120	2.01	18.40

- A. Yes, the employee's exposure is above the OSHA PEL
- B. No, the employee's exposure is not above the OSHA PEL
- C. OSHA does not have a PEL for nuisance dust
- D. The table above does not provide enough information to answer the question

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$$TWA = \frac{(C_1T_1) + (C_2T_2) + \dots (C_nT_n)}{\text{total hours workes}}$$

$$TWA = \frac{12.05 \times 120 + 5.65 \times 240 + 18.40 \times 120}{120 + 240 + 120} = \frac{5010}{480}$$

$$TWA = 10.43 \text{ mg/m}^3$$



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An industrial hygienist collected the following personal dust samples. Is the employee's exposure above the OSHA PEL of **15 mg/m³**?

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